

RAPHAEL RAMOS

Software Engineer | Backend & Data Systems | Python,
Linux, Distributed Telemetry

Campina Grande, PB, Brazil • +55 83 99833-3655
raphaelramosc@gmail.com • linkedin.com/in/raphael-c-1a7430108 • github.com/raphacramos

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (Node.js, React.js), SQL, C/C++, Bash/Shell Script
Technologies & Frameworks: NumPy, Pandas, SciPy, HoloViews, Panel, GNU Radio (SDR/USRP), HDF5, Skyfield, React.js, Node.js, Express, Docker, Git, Linux (SysAdmin)
Knowledge & Domains: Data Structures & Advanced Algorithms, Machine Learning, Statistical Inference, Time Series & Signal Processing, Relational & NoSQL Database Optimization, Clean Architecture

FEATURED ENGINEERING PROJECTS

BINGO Telescope Project — MiniHorn Radio Astronomy Spectrometer Jan 2026 – Present
International Radio Astronomy Software Collaboration
Campina Grande, Brazil

- Python GNU Radio Panel HoloViews HDF5 Skyfield Clean Architecture
- Designed and built an interactive scientific control dashboard and tactical visualization console in Python (Panel, HoloViews, HTML/CSS) for real-time radio spectrum analysis.
 - Architected modular software infrastructure following Clean Architecture principles (Domain, Application, Infrastructure layers), integrating USRP Software-Defined Radio (SDR) hardware for dynamic control of center frequency, gain, and sample rate.
 - Engineered high-throughput data acquisition pipelines streaming real-time SDR RF telemetry directly into structured **HDF5** binary storage with sub-second write cycles and zero sample drop.
 - Integrated the **Skyfield** astronomical library for celestial pointing, ephemeris calculations, and coordinate transformations.

Statistical Inference & Machine Learning Pipeline Aug 2023 – Dec 2023
Applied Quantitative Research
Campina Grande, Brazil

- Python Scikit-Learn NumPy Pandas SciPy Machine Learning
- Developed quantitative analysis models and statistical hypothesis testing pipelines in Python to extract signal patterns and perform predictive modeling on complex datasets.
 - Applied regression and classification algorithms alongside variance analysis and cross-validation to optimize feature selection, variance analysis, and predictive model accuracy.

Inventory Management System & Relational Database Jan 2023 – May 2023
Database Systems Engineering Project
Campina Grande, Brazil

- Python PostgreSQL SQL ACID Git
- Built a transactional backend application in Python backed by PostgreSQL, designing a 3NF normalized schema with optimized indexes and complex queries ensuring ACID compliance.

High-Concurrency E-Commerce REST API & Transactional State Aug 2022 – Dec 2022
Web Systems Engineering Project
Campina Grande, Brazil

- Node.js React.js MongoDB RESTful APIs Docker
- Engineered a decoupled web service architecture with Node.js and MongoDB, implementing high-throughput RESTful endpoints, asynchronous request pipelines, and state synchronization.

TEACHING & ACADEMIC EXPERIENCE

Teaching Assistant — Analysis and Techniques of Algorithms (LEDA / ATAL) Mar 2026 – Aug 2026
Federal University of Campina Grande (UFCG)
Campina Grande, Brazil

- Mentored students in advanced algorithmic problem solving (graph algorithms, dynamic programming, divide-and-conquer, greedy approaches) and rigorous asymptotic time/space complexity analysis ($O(n)$ optimization).

Teaching Assistant — Systems Administration (ADSI) Sep 2025 – Mar 2026
Federal University of Campina Grande (UFCG)
Campina Grande, Brazil

- Guided undergraduates in Linux internals, Bash automation scripting, system process scheduling, network protocol configuration, and infrastructure containerization with Docker.

EDUCATION

Federal University of Campina Grande (UFCG) Aug 2026
Bachelor of Science in Computer Science
Campina Grande, Brazil

- Relevant Coursework:** Data Structures & Algorithms, Analysis and Techniques of Algorithms, Artificial Intelligence, Database Systems, Computer Networks, Information Security, Operating Systems.

University of São Paulo (USP) Jan 2018 – Jun 2019
Undergraduate Studies in Mechatronic Engineering (Transferred)
São Paulo, Brazil

- Scientific Foundation:** Advanced Calculus, General Physics, System Dynamics & Control, Linear Algebra, Digital and Analog Electronics.

LANGUAGES

Portuguese: Native English: Fluent (C2 Proficient) French: Good Working Knowledge Spanish: Good Working Knowledge